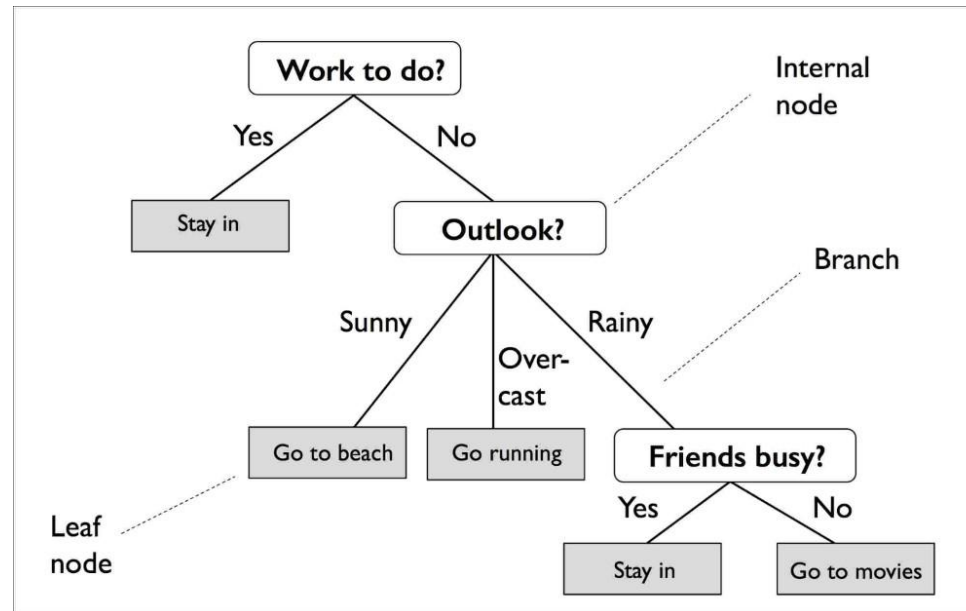


Decision Tree

Decision Trees are a supervised learning method used for both classification and regression tasks. The goal is to create a model that predicts the value of a target variable by learning simple decision rules



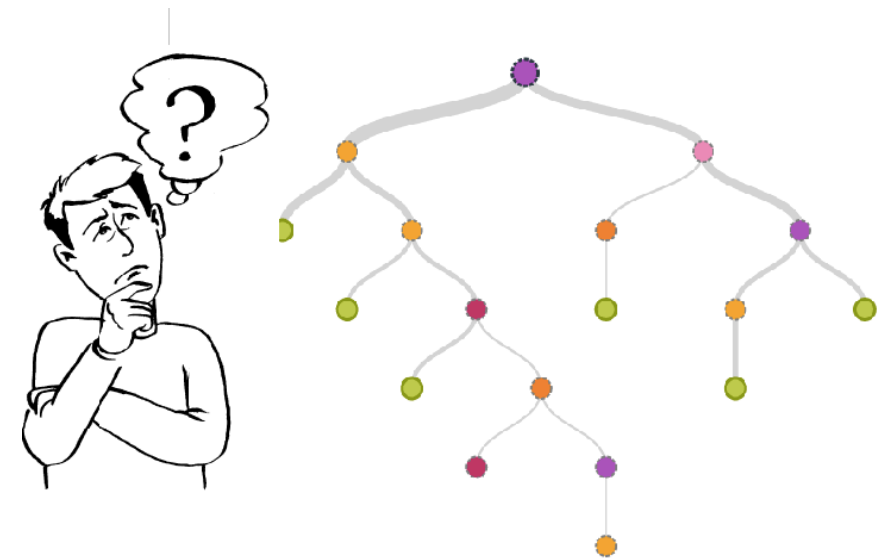
Decision tree is graphical representation of all the possible solutions to a decision, Decision are based on some conditions.



Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

How does tree a decides where to split?

- Gini index/ Gini impurity
- Entropy & Information gain



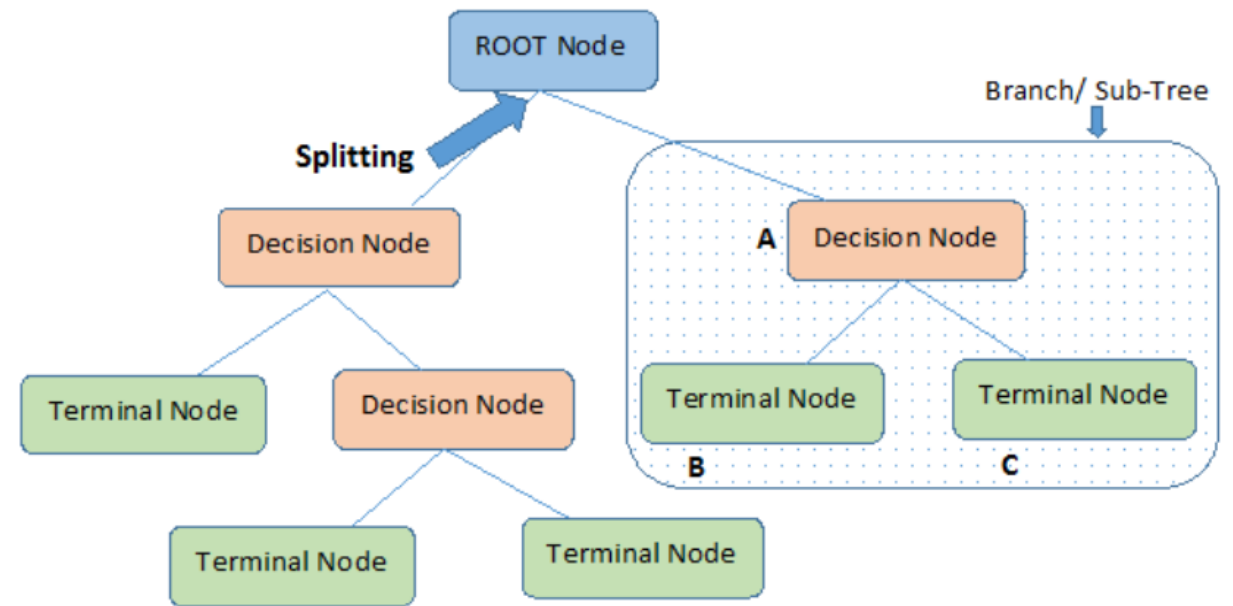
Root Nodes

Decision Nodes

Leaf Nodes

Sub-tree

Pruning: Reduce the complexity and cutting down some nodes to stop overfitting.



1. **Gini index / Gini impurity:** *Favours larger partitions.*
2. The measure of impurity (or purity) used in building decision tree in CART is Gini index

Gini index is measure of inequality in sample.

0 to (1 - 1/n)

Where **n** is the number of classes.

$$Gini\ index = 1 - \sum_{i=1}^n p_i^2$$



Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No



$$\text{Gini index} = 1 - \sum_{i=1}^n p_i^2$$

Outlook

Outlook	Yes	No	# Instances
sunny	2	3	5
overcast	4	0	4
rainfall	3	2	5

Temperature

Temperature	Yes	No	# Instances
hot	2	2	4
cool	3	1	4
mild	4	2	6

Gini index (outlook=sunny)= $1 - ((2/5)^2 + (3/5)^2) = 1 - 0.16 - 0.36 = 0.48$

Gini index(outlook=overcast)= $1 - (4/4)^2 - (0/4)^2 = 1 - 1 - 0 = 0$

Gini index(outlook=rainfall)= $1 - (3/5)^2 - (2/5)^2 = 1 - 0.36 - 0.16 = 0.48$

$(0.48 + 0 + 0.48) / 3 = 0.32$

Now , we will calculate the weighted sum of Gini index for outlook features,

Gini(outlook) = $(5/14) * 0.48 + (4/14) * 0 + (5/14) * 0.48 = 0.342$

Gini(temperature=hot) = $1 - (2/4)^2 - (2/4)^2 = 0.5$

Gini(temperature=cool) = $1 - (3/4)^2 - (1/4)^2 = 0.375$

Gini(temperature=mild) = $1 - (4/6)^2 - (2/6)^2 = 0.445$

Now, the weighted sum of Gini index for temperature features can be calculated as,

Gini(temperature)= $(4/14) * 0.5 + (4/14) * 0.375 + (6/14) * 0.445 = 0.439$



Humidity

Humidity	Yes	No	# Instances
high	3	4	7
Normal	6	1	7

Humidity is a binary class feature , it can take two value high and normal.

$$\text{Gini}(\text{humidity}=\text{high}) = 1 - (3/7)^2 - (4/7)^2 = 0.489$$

$$\text{Gini}(\text{humidity}=\text{normal}) = 1 - (6/7)^2 - (1/7)^2 = 0.244$$

Now, the weighted sum of Gini index for humidity features can be calculated as,

$$\text{Gini}(\text{humidity}) = (7/14) * 0.489 + (7/14) * 0.244 = 0.367$$

Wind

wind	Yes	No	# Instances
weak	6	2	8
strong	3	3	6

wind is a binary class feature , it can take two value weak and strong.

$$\text{Gini}(\text{wind}=\text{weak}) = 1 - (6/8)^2 - (2/8)^2 = 0.375$$

$$\text{Gini}(\text{wind}=\text{strong}) = 1 - (3/6)^2 - (3/6)^2 = 0.5$$

Now, the weighted sum of Gini index for wind features can be calculated as,

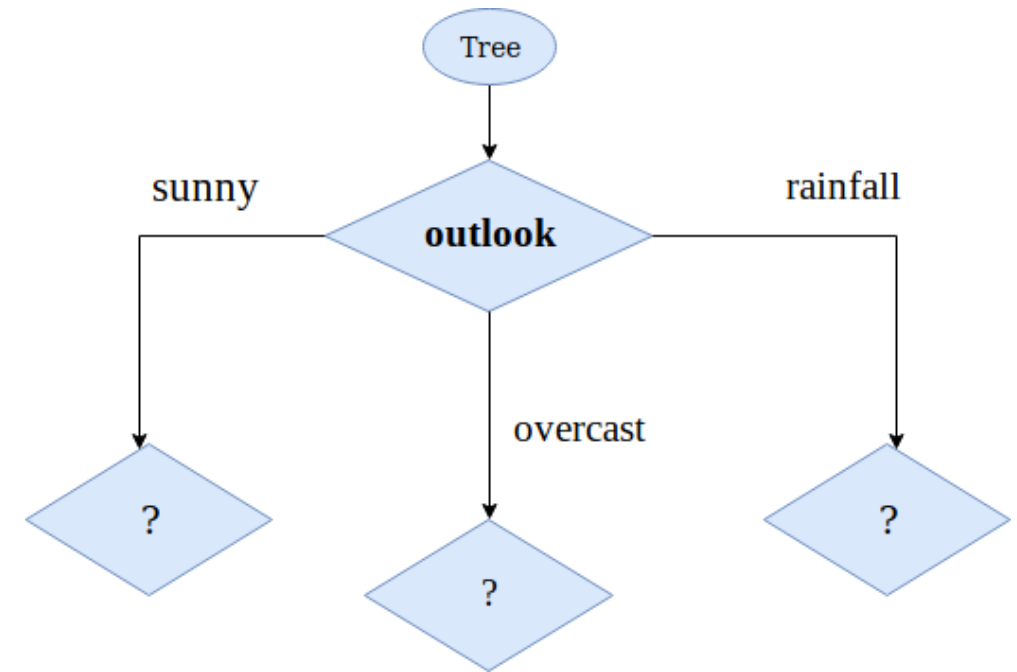
$$\text{Gini}(\text{wind}) = (8/14) * 0.375 + (6/14) * 0.5 = 0.428$$



Decision for root node

Features	Gini Index
outlook	0.342
temperature	0.439
humidity	0.367
wind	0.428

From table, you can see that Gini index for outlook feature is lowest. So we get our root node.

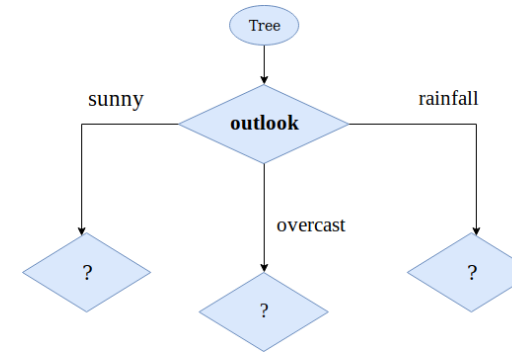


Lets calculate the Gini index on sub data set for outlook feature, as you can see we have three sub data section, sunny, overcast, and rainfall of outlook feature. we will use same method as above to find next split.



Gini index for temperature on sunny outlook

Day	outlook	temperature	humidity	wind	decision
1	sunny	hot	high	weak	No
2	sunny	hot	high	strong	No
8	sunny	mild	high	weak	No
9	sunny	cool	normal	weak	Yes
11	sunny	mild	normal	strong	Yes



Temperature	Yes	No	# Instances
hot	0	2	2
cool	1	1	1
mild	1	1	2

$$\text{Gini}(\text{outlook}=\text{sunny} \ \& \ \text{temperature}=\text{hot}) = 1 - (0/2)^2 - (2/2)^2 = 0$$

$$\text{Gini}(\text{outlook}=\text{sunny} \ \& \ \text{temperature}=\text{cool}) = 1 - (1/1)^2 - (0/1)^2 = 0$$

$$\text{Gini}(\text{outlook}=\text{sunny} \ \& \ \text{temperature}=\text{mild}) = 1 - (1/2)^2 - (1/2)^2 = 0.5$$

Now, the weighted sum of Gini index for temperature on sunny outlook features can be calculated as,

$$\text{Gini}(\text{outlook}=\text{sunny} \ \& \ \text{temperature}) = (2/5) * 0 + (1/5) * 0 + (2/5) * 0.5 = 0.2$$



Gini Index for humidity on sunny outlook

Humidity	Yes	No	# Instances
high	0	3	3
Normal	2	0	2

$$\text{Gini}(\text{outlook}=\text{sunny} \ \& \ \text{humidity}=\text{high}) = 1 - (0/3)^2 - (3/3)^2 = 0$$

$$\text{Gini}(\text{outlook}=\text{sunny} \ \& \ \text{humidity}=\text{normal}) = 1 - (2/2)^2 - (0/2)^2 = 0$$

Now, the weighted sum of Gini index for humidity on sunny outlook features can be calculated as,

$$\text{Gini}(\text{outlook} = \text{sunny} \ \& \ \text{humidity}) = (3/5) * 0 + (2/5) * 0 = 0$$

Gini Index for wind on sunny outlook

wind	Yes	No	# Instances
weak	1	2	3
strong	1	1	2

$$\text{Gini}(\text{outlook}=\text{sunny} \ \& \ \text{wind}=\text{weak}) = 1 - (1/3)^2 - (2/3)^2 = 0.44$$

$$\text{Gini}(\text{outlook}=\text{sunny} \ \& \ \text{wind}=\text{strong}) = 1 - (1/2)^2 - (1/2)^2 = 0.5$$

Now, the weighted sum of Gini index for wind on sunny outlook features can be calculated as,

$$\text{Gini}(\text{outlook} = \text{sunny} \ \& \ \text{wind}) = (3/5) * 0.44 + (2/5)$$

$$* 0.5 = 0.266 + 0.2 = 0.466$$



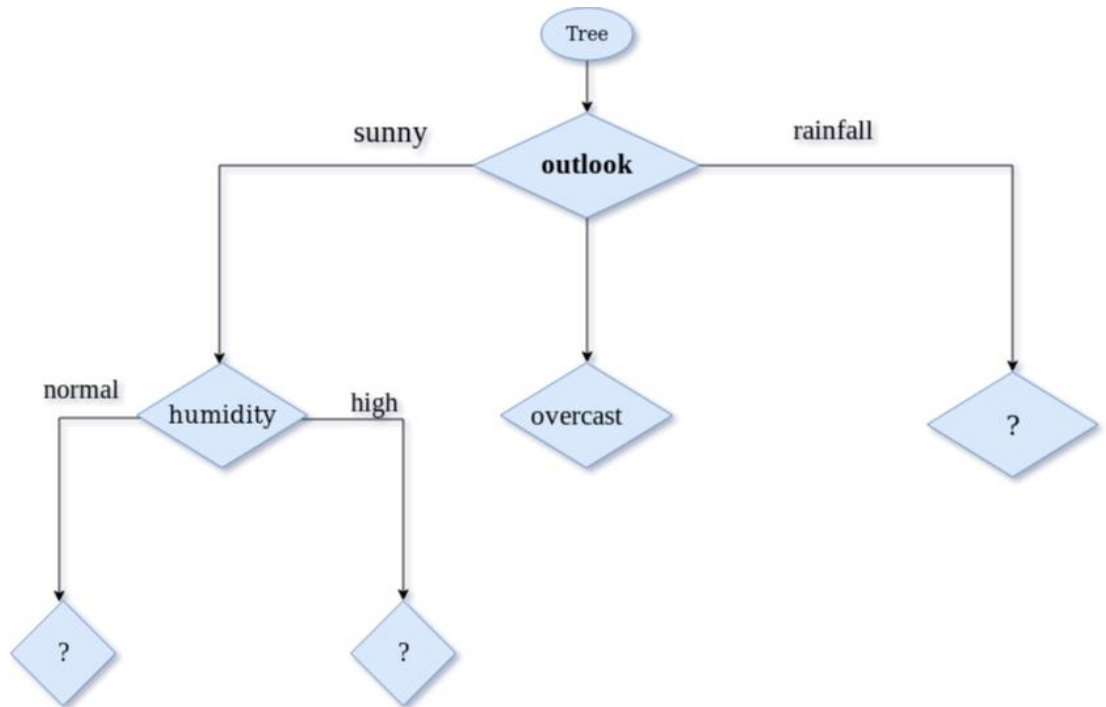
Decision on sunny outlook factor

Features	Gini Index
temperature	0.2
humidity	0
wind	0.466

we have calculated the Gini index of all the features when the outlook is sunny. You can infer that humidity has lowest value. so next node will be humidity.

Now, Lets focus on sub data for overcast outlook feature.

Day	outlook	temperature	humidity	wind	decision
3	overcast	hot	high	weak	Yes
7	overcast	cool	normal	strong	Yes
12	overcast	mild	high	strong	Yes
13	overcast	hot	normal	weak	Yes

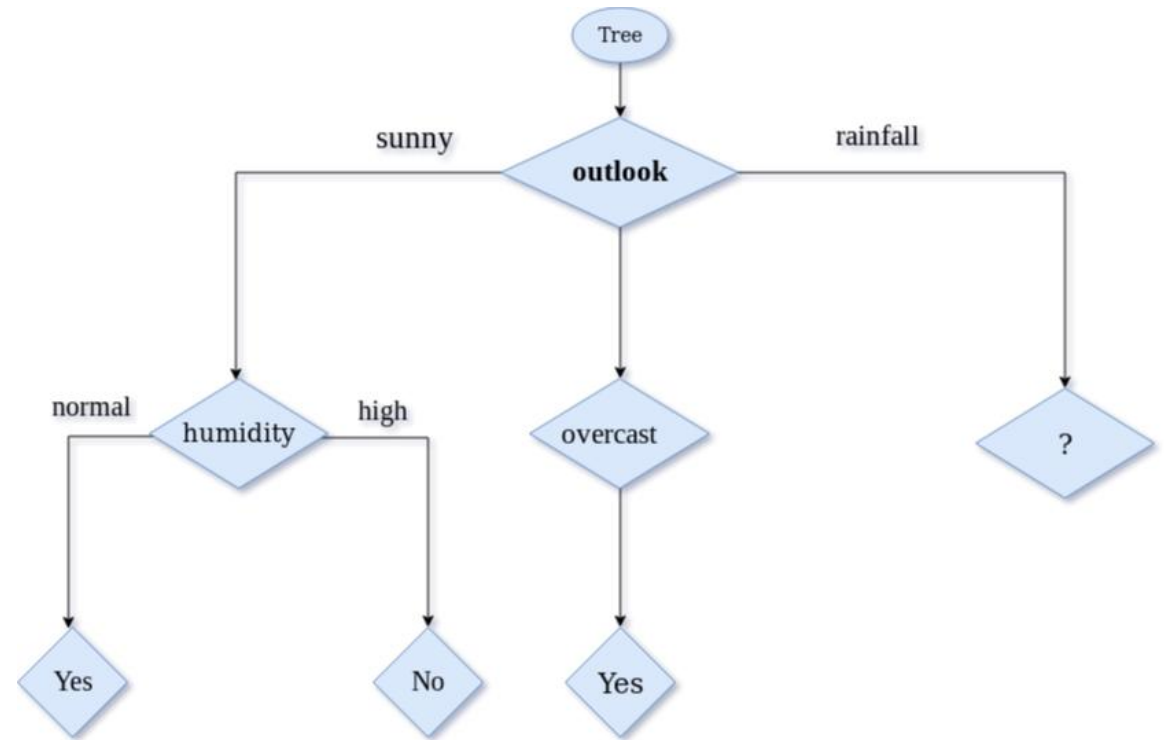


Now, Lets focus on sub data for high and normal humidity feature.

Day	outlook	temperature	humidity	wind	decision
1	sunny	hot	high	weak	No
2	sunny	hot	high	strong	No
8	sunny	mild	high	weak	No

Day	outlook	temperature	humidity	wind	decision
9	sunny	cool	normal	weak	Yes
11	sunny	mild	normal	strong	Yes

From the given two table, the decision is always 'No' when humidity is 'high' and decision is always 'Yes' when humidity is 'normal'. So we got leaf node. now decision tree can be viewed as,



Now, Lets focus on sub data for rainfall outlook feature. we need to find the Gini index for temperature, humidity and wind feature respectively.

Day	outlook	temperature	humidity	wind	Decision
4	rain	mild	high	weak	Yes
5	rain	cool	normal	weak	Yes
6	rain	cool	normal	strong	No
10	rain	mild	normal	weak	Yes
14	rain	mild	high	strong	No

Gini index for temperature for rainfall outlook

temperature	Yes	No	# Instances
cool	1	1	2
mild	2	1	3

$$\text{Gini}(\text{outlook}=\text{rainfall and temp.}=\text{Cool}) = 1 - (1/2)^2 - (1/2)^2 = 0.5$$

$$\text{Gini}(\text{outlook}=\text{rainfall and temp.}=\text{Mild}) = 1 - (2/3)^2 - (1/3)^2 = 0.444$$

$$\text{Gini}(\text{outlook}=\text{rainfall and temp.}) = (2/5)*0.5 + (3/5)*0.444 = 0.466$$



Gini index for humidity for rainfall outlook

humidity	Yes	No	# Instances
high	1	1	2
normal	2	1	3

$$\text{Gini}(\text{outlook}=\text{rainfall and humidity}=\text{high}) = 1 - (1/2)^2 - (1/2)^2 = 0.5$$

$$\text{Gini}(\text{outlook}=\text{rainfall and humidity}=\text{normal}) = 1 - (2/3)^2 - (1/3)^2 = 0.444$$

$$\text{Gini}(\text{Outlook}=\text{rainfall and humidity}) = (2/5)*(0.5) + (3/5)*0.444 = 0.466$$

Gini index for wind for rainfall outlook feature

wind	Yes	No	# Instances
weak	3	0	3
strong	0	2	2

$$\text{Gini}(\text{outlook}=\text{rainfall and wind}=\text{weak}) = 1 - (3/3)^2 - (0/3)^2 = 0$$

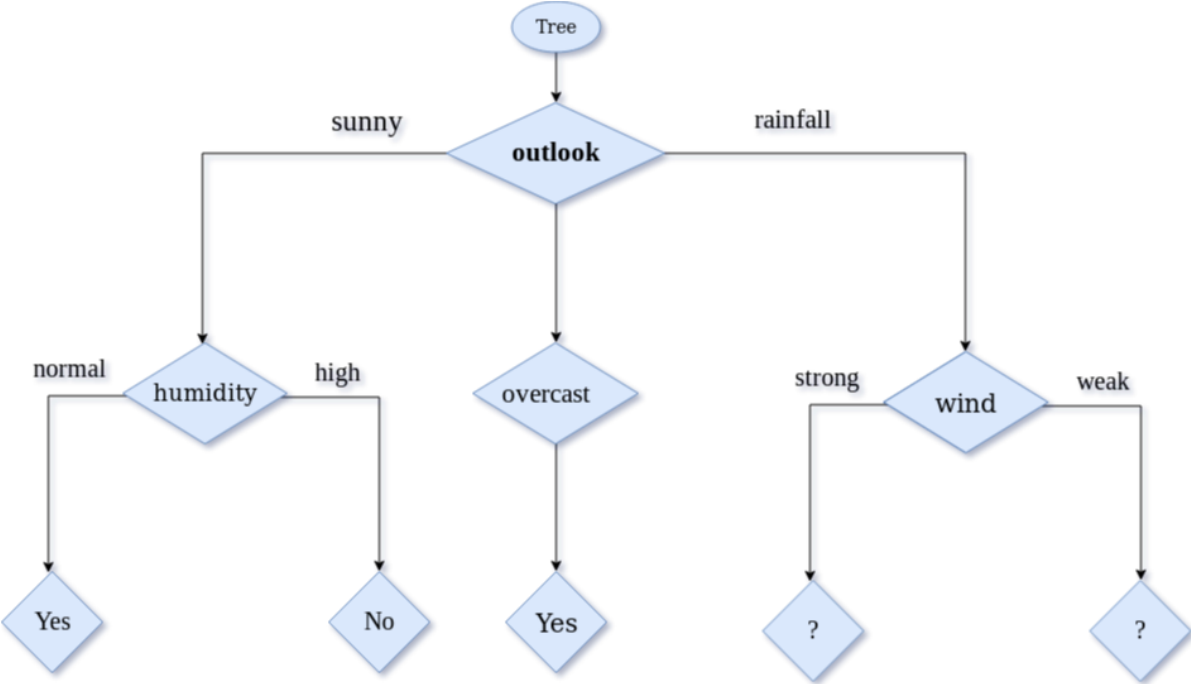
$$\text{Gini}(\text{outlook}=\text{rainfall and wind}=\text{strong}) = 1 - (0/2)^2 - (2/2)^2 = 0$$

$$\text{Gini}(\text{outlook}=\text{rainfall and wind}) = (3/5)*0 + (2/5)*0 = 0$$



Decision on rainfall outlook factor

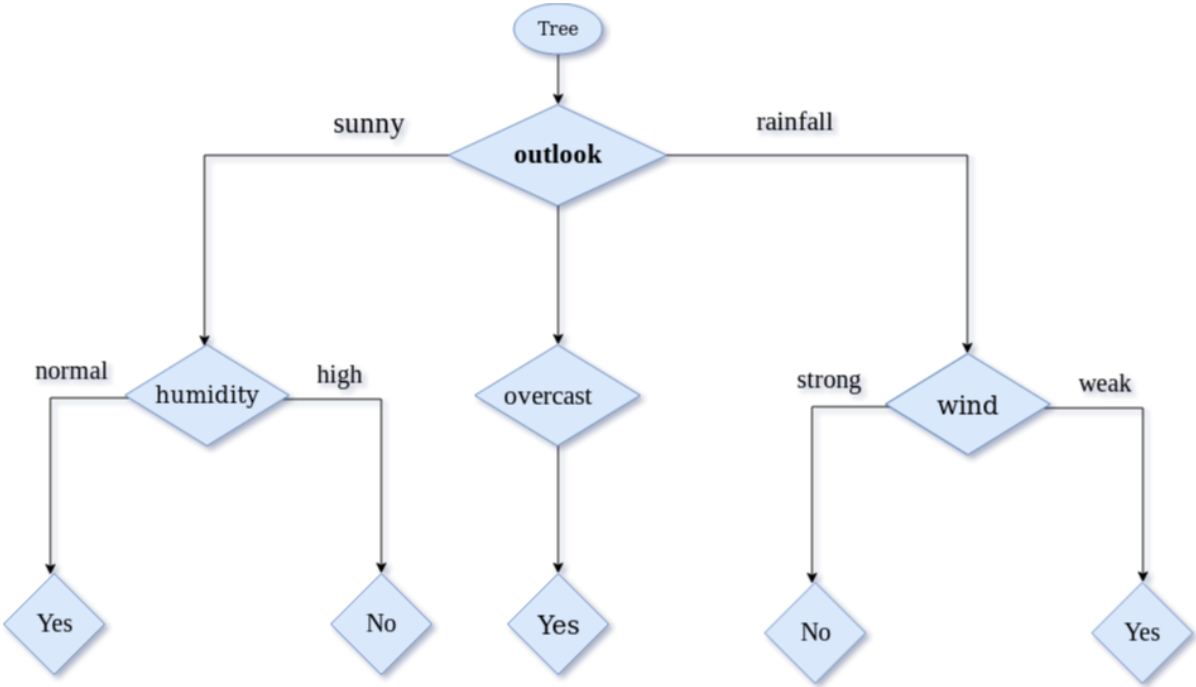
Features	Gini Index
temperature	0.466
humidity	0.466
wind	0



Now, Lets focus on sub data strong and weak for wind rainfall feature.

Day	outlook	temperature	humidity	wind	decision
6	rainfall	cool	normal	strong	No
14	rainfall	mild	high	strong	No

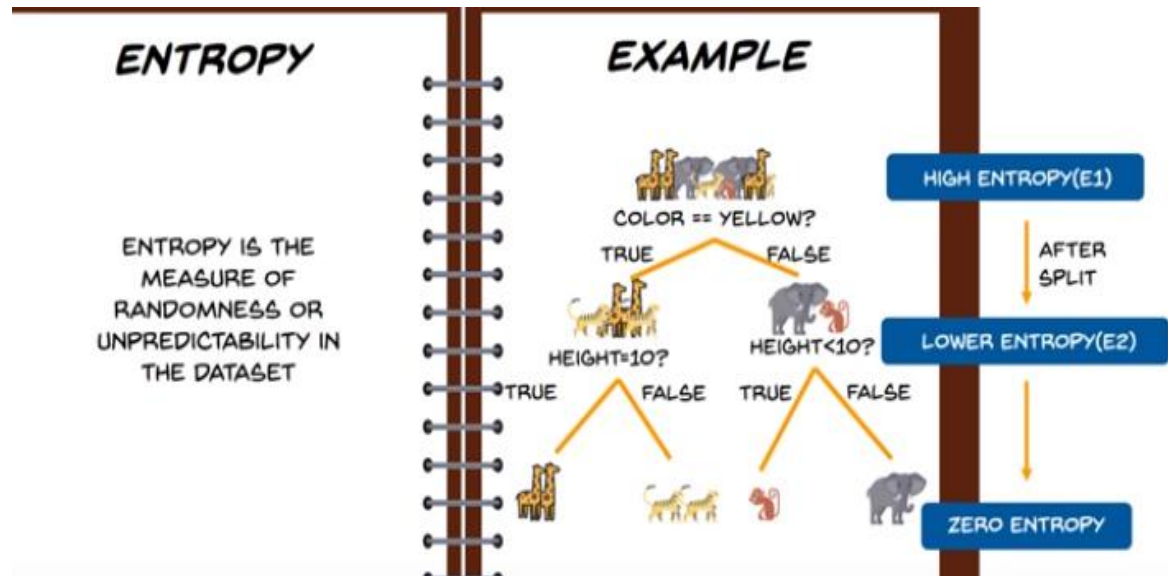
Day	outlook	temperature	humidity	wind	decision
4	rainfall	mild	high	weak	Yes
5	rainfall	cool	normal	weak	Yes
10	rainfall	mild	normal	weak	Yes



From the above two table, the decision is always 'No' when wind is 'strong' and decision is always 'Yes' when wind is 'weak'. So we got leaf node.

Entropy

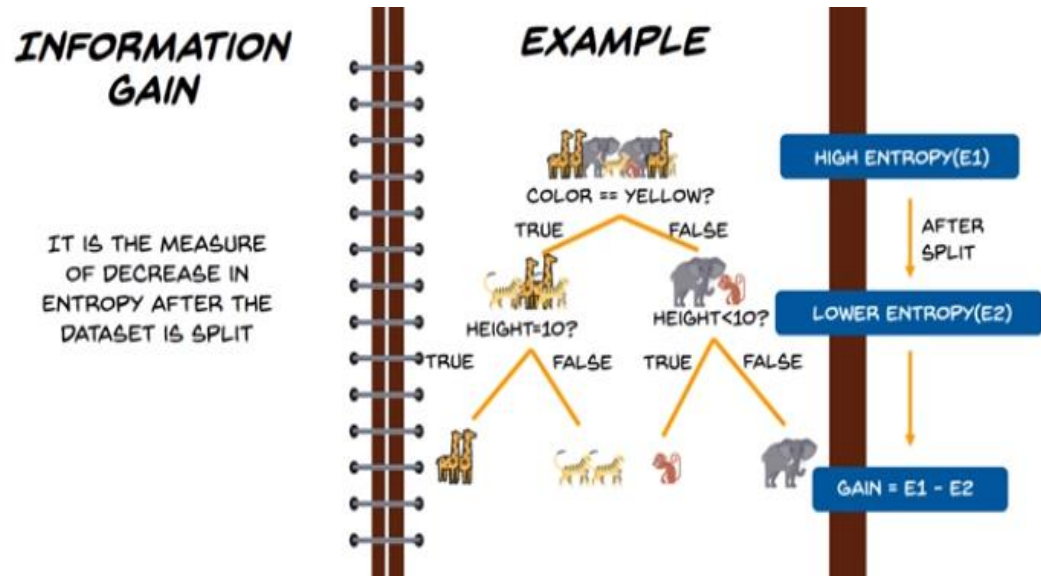
Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes



$$Entropy = - \sum_{i=1}^n p_i * \log(p_i)$$

Where: P_i is a fraction of example in given class





Information gain (IG): measures how much “information” a feature gives us about the class.

Information gain measures the reduction of uncertainty given some feature and it is also a deciding factor for which attribute should be selected as a decision node or root node.

$$Entropy(S) = \sum_{i=1}^c -p_i \log_2 p_i$$

$$Gain(S, A) = Entropy(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v)$$

◆ **S** = The full dataset (or current subset) at a node

◆ **A** = The attribute/feature you are evaluating to split on

S_v = Subset of **S** where attribute **A** has value **v**





Step 1: Calculate Entropy(S)

- Total records = 14
- PlayTennis = Yes : 9
- PlayTennis = No : 5

$$Entropy(S) = - \left(\frac{9}{14} \right) \log_2 \left(\frac{9}{14} \right) - \left(\frac{5}{14} \right) \log_2 \left(\frac{5}{14} \right)$$

$$Entropy(S) \approx -0.643 \cdot \log_2(0.643) - 0.357 \cdot \log_2(0.357) \approx 0.940$$

Step-2

♦ Attribute: Outlook

- Sunny: 5 samples → [2 Yes, 3 No]

$$Entropy = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} \approx 0.971$$

- Overcast: 4 samples → [4 Yes, 0 No]

$$Entropy = 0$$

- Rain: 5 samples → [3 Yes, 2 No]

$$Entropy = -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} \approx 0.971$$

Weighted entropy:

$$\frac{5}{14} \cdot 0.971 + \frac{4}{14} \cdot 0 + \frac{5}{14} \cdot 0.971 \approx 0.694$$

$$Gain(S, Outlook) = 0.940 - 0.694 = 0.246$$



◆ Attribute: Temperature

- Hot: 4 samples → [2 Yes, 2 No] → Entropy = 1.0
- Mild: 6 samples → [4 Yes, 2 No] → Entropy ≈ 0.918
- Cool: 4 samples → [3 Yes, 1 No] → Entropy ≈ 0.811

$$Weighted = \frac{4}{14} \cdot 1 + \frac{6}{14} \cdot 0.918 + \frac{4}{14} \cdot 0.811 \approx 0.911$$

$$Gain(S, Temperature) = 0.940 - 0.911 = 0.029$$

◆ Attribute: Humidity

- High: 7 samples → [3 Yes, 4 No] → Entropy ≈ 0.985
- Normal: 7 samples → [6 Yes, 1 No] → Entropy ≈ 0.591

$$Weighted = \frac{7}{14} \cdot 0.985 + \frac{7}{14} \cdot 0.591 = 0.788$$

$$Gain(S, Humidity) = 0.940 - 0.788 = 0.152$$

◆ Attribute: Wind

- Weak: 8 samples → [6 Yes, 2 No] → Entropy ≈ 0.811
- Strong: 6 samples → [3 Yes, 3 No] → Entropy = 1

$$Weighted = \frac{8}{14} \cdot 0.811 + \frac{6}{14} \cdot 1 \approx 0.892$$

$$Gain(S, Wind) = 0.940 - 0.892 = 0.048$$



✓ Final Information Gain Table

Attribute	Information Gain
Outlook	0.246
Temperature	0.029
Humidity	0.152
Wind	0.048

Outlook has the highest Information Gain (0.246) → becomes the **root node** of the decision tree.



ID3 (Iterative Dichotomiser 3)

Used For:

- Building decision trees from categorical data

✓ How it works:

- Splits the data using Information Gain
- Selects the feature that gives the **highest Information Gain** at each step

✓ Key Concepts:

- Entropy: Measures impurity or randomness in the data

$$Entropy(S) = - \sum p_i \log_2(p_i)$$

- Information Gain: Reduction in entropy after splitting

$$Gain(S, A) = Entropy(S) - \sum \frac{|S_v|}{|S|} \cdot Entropy(S_v)$$

✓ Limitations:

- Handles **categorical features only**
- Can **overfit** the training data
- Bias towards features with **more categories**



2. C4.5 (Successor of ID3)

✓ Improvements over ID3:

- Handles **both categorical and continuous** attributes
- Uses **Gain Ratio** instead of just Information Gain
- Performs **pruning** to avoid overfitting
- Handles **missing values**

✓ How it works:

- Gain Ratio = Information Gain / Intrinsic Value

$$GainRatio(A) = \frac{InformationGain(A)}{SplitInformation(A)}$$

- Prunes tree after building (post-pruning)

✓ Advantages:

- More accurate and robust than ID3
- Suitable for real-world data with mixed types



3. **C5.0 (Commercial version of C4.5)**

✓ Developed by:

- RuleQuest Research, based on C4.5

✓ Key Improvements over C4.5:

- Faster and more memory-efficient
- Boosting support (ensemble of trees)
- Smaller decision trees, better accuracy
- Supports weights for training instances
- Handles large datasets better

✓ Features:

- Uses entropy and gain ratio, like C4.5
- Performs error-based pruning



- Commercial and proprietary, but some free versions exist (e.g., C50 package in R)

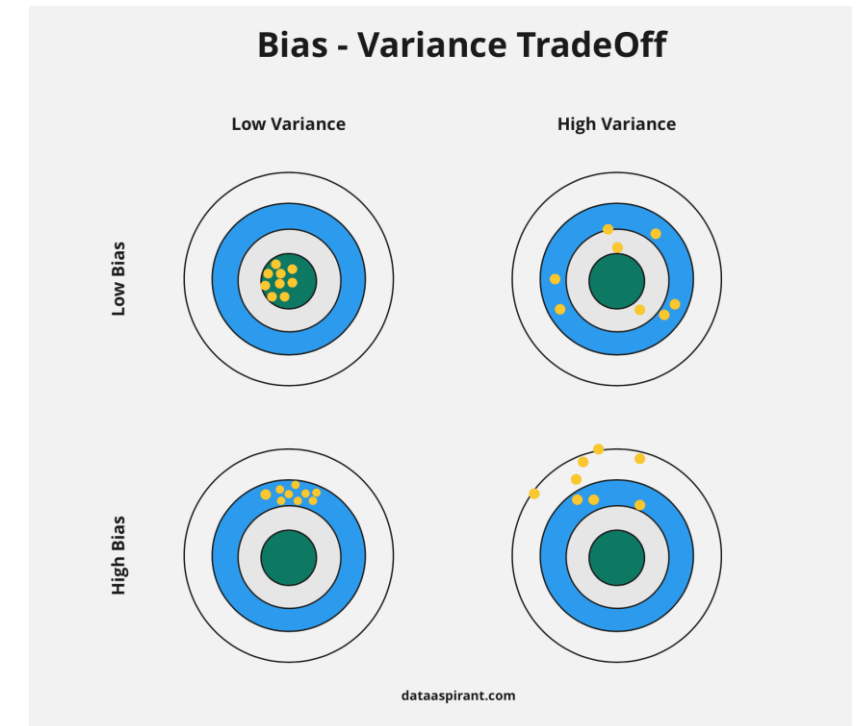
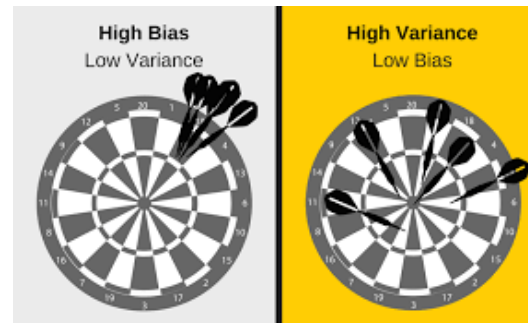
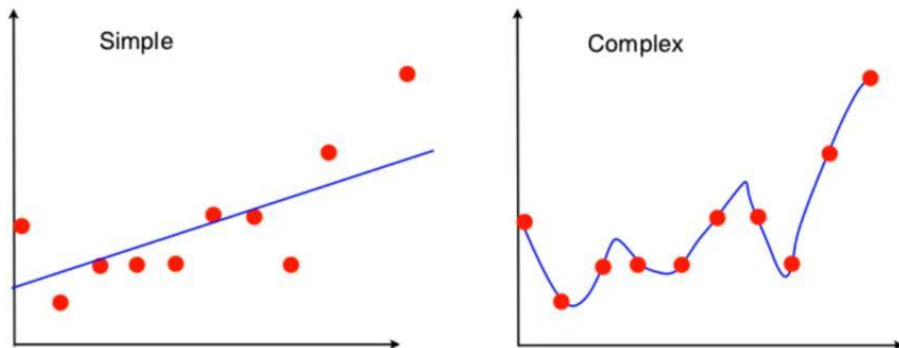


Bias / Error/ Error due to bias

Bias is the difference between the average prediction of our model and the correct value which we are trying to predict.

Variance

Variance is the variability of model prediction for a given data point or a value which tells us spread of our data.



Overfitting & Underfitting

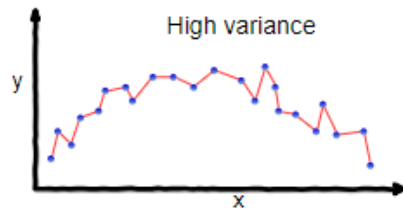
Overfitting

Overfitting happens when a model learns the detail and noise in the training data to the extent that it negatively impacts the performance of the model on new data.

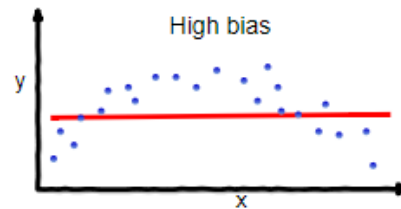
Good performance on the training data, poor generalization to other data.

Underfitting

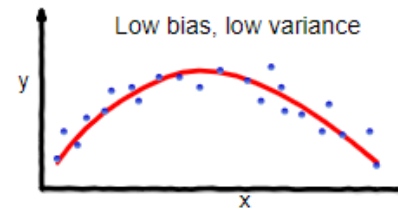
Poor performance on the training data and poor generalization to other data



overfitting



underfitting



Good balance





A

Not interested in learning

Class test ~50%

Test ~47%

Under-fit/ biased learning



B

Memorizing the lessons

Class test ~98%

Test ~69%

Over-fit/ Memorizing



C

Conceptual Learning

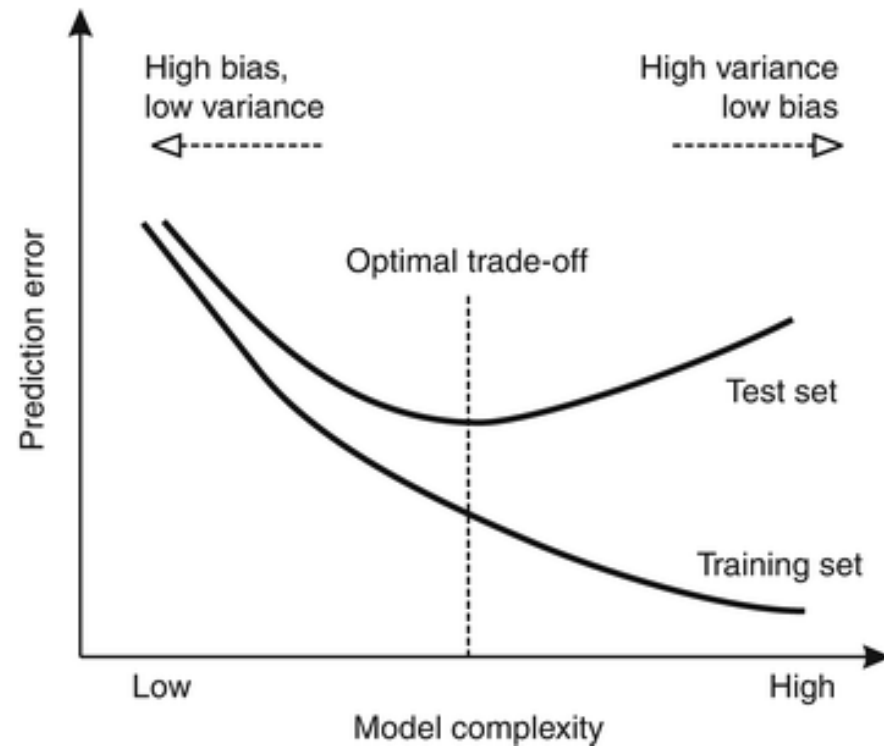
Class test ~92%

Test ~89%

Best-fit

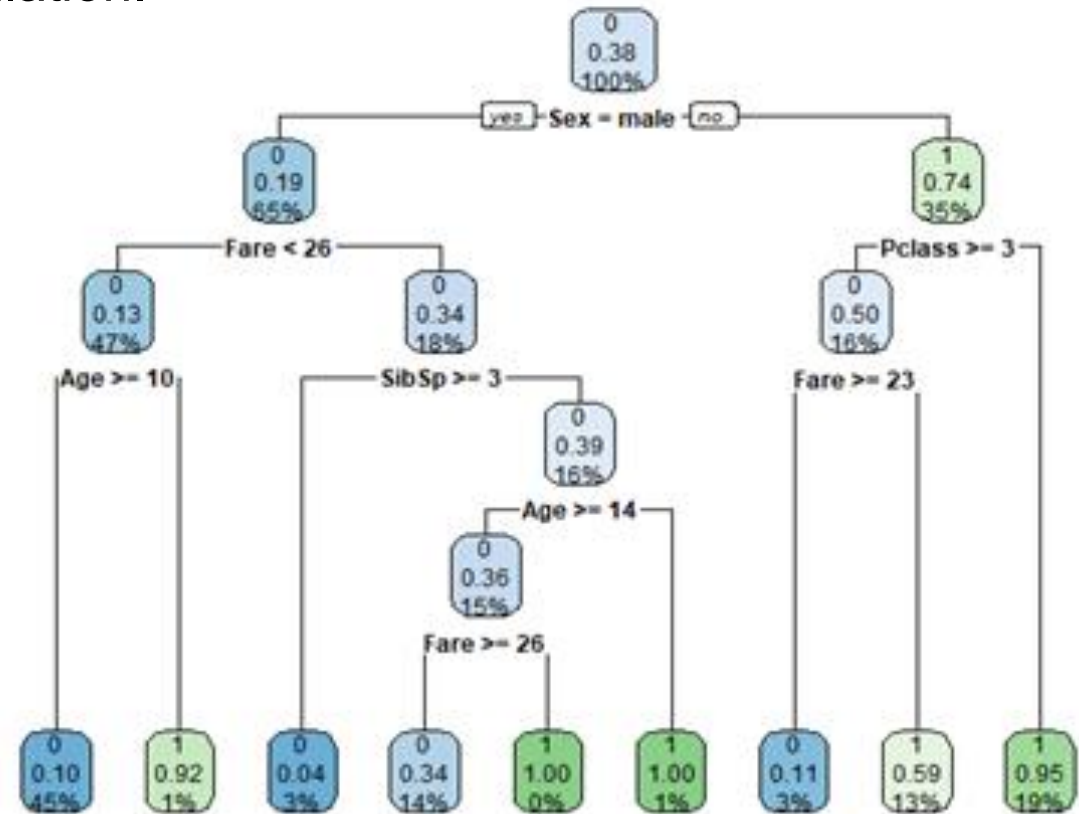


Trade-off



Reduction in Variance

Reduction in Variance is an algorithm use for continuous target variable(Regression Problem). The split with lower variance is selected as the criteria to split the population.



Mse, mae



Question: If you are not using Standard Scaler before modeling, so what will be effect in modeling.....

For KNN, Kmean where we are using distance formula: standardization is important

But for SVM, DT, RF these algorithm standardization is not important

What is different between Entropy and Gini Impurity

Entropy takes more computational time as compared with gini impurity

Gini impurity is many times faster than entropy

Question: If you have continuous features, so what process we have to follow for Decision Tree

- 1) Sorting all the value
- 2) Select one threshold value
- 3) Followed as next condition and make branches
- 4) Then find Information Gain/ Gini Index

Time Complexity increasing with numerical features.



Chi square

It is algorithm to find out the statistical significance between the difference between sub-node and parent node.

Split on Performance in Class

